Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7 FT **1** HT	(a)	(i)		Any 2 × (1) from: Increases after 1990 or by 1995 (1) Drops in 2000 (1) Rises by 2005/2010 (1) Remains approximately steady after 2010 (1) Each marking point must include a relevant year Increases, decreases and increases (1 mark only)		2			3	
		(ii)		Scales (x axis 5 years per 2 cm square, y axis 50 per 2 cm square) (1) 7 correct plots (within appropriate small square) (2) 6 correct plots (within appropriate small square) (1) 5 correct plots (0) curve of best fit / joining points with curve / point to point (1)		4			4	
		(iii)		CO_2 emissions decrease (over time)/negative correlation (1) Rate of decrease in the last 10 years is greater than the first 10 years / smaller decrease before 2005 than after 2005 (1) Ignore reference to plateau. Allow ecf from incorrect plot		2				

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		Graph 1 shows an increase over time (1) so does not correlate with drawn graph (1) OR My graph shows emissions (from UK) decrease (1) but atmospheric concentration is increasing (1)			2			
		(v)		Rate of photosynthesis varies throughout the year (1) Uptake of CO_2 varies (1) Lower rate (in November) corresponds with peak / Higher rate (in May) correlates with dip (1) Accept: Rate of decay varies throughout the year (1) CO_2 release varies (1) higher rate (of decay/respiration of decomposers) corresponds with peak / lower rate (of decay/respiration of decomposers) corresponds with dip (1)			3			

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks Available					
	(b)			288 000 / **68 520** / 11 880 / 368 400 / 72.37 17 808 / 12 117 / **6 720** / 36 645 / **7.20/7.1986/ 7.19/7.00** 3 × (1)		3			3	(b)
	(c)			Water vapour becomes largest factor (1) All others decrease (1)		2				(c)
	(d)			man made contributions are very small / natural contributions are high			1			(d)
	(e) HT Only			**Indicative content** In graph 2, variations in the sunspot cycle length and the temperature anomaly are closely linked. This is true for the whole time scale. Between 1910 and 1970 the CO_2 concentration gradually increases but during this time the temperature anomaly increases steeply, peaks and falls. In graph 3, there is a close match between the increasing trends of CO_2 concentration and temperature anomaly. The steepest increase in temperature since 1960 is matched by a fluctuating but steady sunspot activity. The evidence is not conclusive since graph 2 supports solar activity being responsible for global warming but graph 3 supports the CO_2 cause. **5-6 marks** Comprehensive comparison of trends of the 3 variables in both graphs. A statement that the evidence is inconclusive is required. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.*						

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				3-4 marks Comparison of trends of at least two variables in both graphs. May not state that evidence in graphs contradicts each other. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1-2 marks** Limited comparisons made of trends or some analysis of one graph. No conclusions made. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** No attempt made or no response worthy of credit.						
				Section B total	**3**	**14**	**8**	**25**	**12**	**0**